

DDA5001 Machine Learning

Training versus Testing (Part I)

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Recap: Solution of Least Squares

LS formulation:

$$\min_{\boldsymbol{\theta}} \frac{1}{n} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2.$$

- ▶ The optimal solution $\hat{\boldsymbol{\theta}}$ satisfies

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{y}.$$

- ▶ Case I: $\mathbf{X} \in \mathbb{R}^{n \times d}$ has full column rank, then

$$\hat{\boldsymbol{\theta}} = \left(\mathbf{X}^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{X}^\dagger \mathbf{y}.$$

- ▶ Case II: $\mathbf{X} \in \mathbb{R}^{n \times d}$ does not have full column rank. The typical case is, $n < d$. It means **overfitting**. LS has **infinitely many** solutions.

What We Have Learned about Supervised Learning

- ▶ Components of supervised learning.
- ▶ Linear classification and the perceptron algorithm.
- ▶ Linear regression and least squares.

Next:

- ▶ Let us take linear classification as example. The perceptron algorithm is learned based on the training dataset.
- ▶ How much it can say about the test dataset?
- ▶ The same question applies to linear regression as well.

↪ This question is about **training versus testing** (also known as **generalization**).

Agenda

Feasibility of Learning

Several Useful Probabilistic Inequalities

Setup for General Training versus Testing

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job	salary	approve ?
3	10	-1
0	0	+1

Test		
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- ▶ The **(really) unknown** target function g is the object of learning. Learning is all about to infer g **outside of the seen training dataset**.
- ▶ There are extreme cases, e.g., we know g on the training dataset, but nothing on other unseen (test) data points. Learning in this case is obviously infeasible.
- ▶ Know things we have already seen, this is not learning, it is **memorizing**.

When and How?

From the previous example, we know that learning is not feasible if we

- ▶ make no assumptions about the connections between the training and test data points, and
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Fortunately, learning will be possible if we

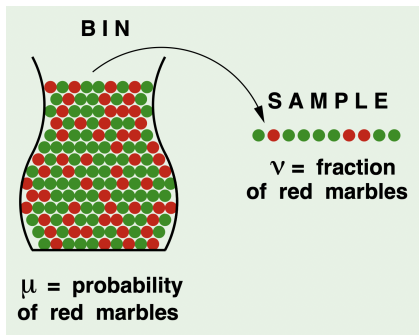
- ▶ make some assumptions that the training and testing data are related in some way. The most common assumption:

The training and test data are independent and identically distributed (i.i.d.)

- ▶ predict something about the test data **in a probabilistic way**.

A Bin Sampling Example: i.i.d. and Randomness

- ▶ Consider a bin with **red** and **green** marbles.
 - $\Pr[\text{red marble}] = \mu$.
 - $\Pr[\text{green marble}] = 1 - \mu$.
- ▶ The learning task is to learn μ .
- ▶ μ is **unknown** to us.
- ▶ We pick n marbles **randomly** and **independently** (with **replacement**). This means i.i.d.



The **fraction** of red marbles is ν .

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- ▶ **Yes.** By intuition, sample frequency ν is **likely** close to the bin frequency μ , once n is not too small.

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It is “**Possible** versus **Probable**”.

- ▶ We choose the “probable” characterization to describe feasibility of learning: learning is feasible in a **probabilistic** sense
 - ▶ The sample frequency ν is close to μ **with high probability**.
- ▶ We need several useful probabilistic inequalities to obtain the “probable” results.

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Random Variable

Suppose X is a random variable, how to quantify the behavior of X ?

- ▶ Probability density function $p(x)$

In principle a random variable X can possibly take any value in $(-\infty, +\infty)$

Random Variable

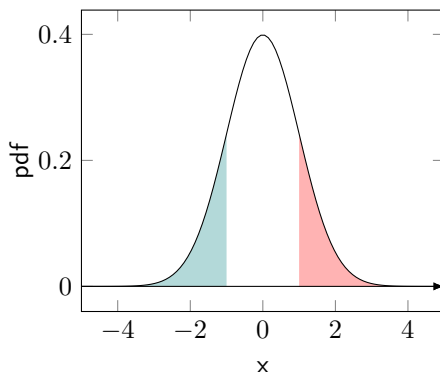
Suppose X is a random variable, how to quantify the behavior of X ?

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We can say something like

$$\Pr[X \geq t]$$



Concentration Inequality for Sub-Gaussian

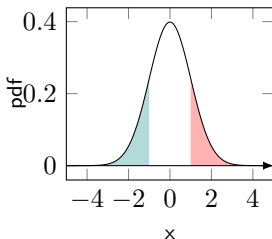
Theorem: Sub-Gaussian Concentration

Suppose X is a **sub-Gaussian** random variable with mean μ and parameter σ , then for any $t > 0$, we have

$$\Pr[|X - \mu| \geq t] \leq 2e^{-\frac{t^2}{2\sigma^2}}$$

- ▶ Sub-Gaussian distributions include Gaussian distributions and any **bounded** distribution.
- ▶ Tail probability decays exponentially with respect to t .
- ▶ Equivalently,

$$\Pr[|X - \mu| \leq t] \geq 1 - 2e^{-\frac{t^2}{2\sigma^2}}$$



Hoeffding's Inequality

- **Fact:** Any bounded random variable on $[a, b]$ are sub-Gaussian random variable with sub-Gaussian parameter $\sigma \leq \frac{b-a}{2}$.
- We can have the following corollary of Hoeffding's inequality for bounded random variables.

Corollary: Hoeffding's inequality for bounded random variables

Suppose X_i are **independent** random variables with mean μ_i and bounded on $[a_i, b_i]$ for $i = 1, \dots, n$, then for any $t > 0$, we have

$$\Pr \left[\sum_{i=1}^n (X_i - \mu_i) \geq t \right] \leq e^{-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}}$$

and

$$\Pr \left[\left| \sum_{i=1}^n (X_i - \mu_i) \right| \geq t \right] \leq 2e^{-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}}$$

Back to Bin Sampling Example

Example: In the bin sampling example, the sampled red marble follows Binomial- (n, μ) . What is lower bound of $\Pr[|\nu - \mu| \leq t]$?

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Suppose that X is the total number of red marbles out of n samples. Each sample is sub-Gaussian with parameter $\sigma \leq \frac{b-a}{2} = \frac{1}{2}$ as each sample can either take value 1 (with prob. μ) or take value 0 (with prob. $1 - \mu$).

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Hence, by Hoeffding's inequality,

$$\Pr[|\nu - \mu| \geq t] = \Pr[|X - \mu n| \geq nt] \leq 2e^{-2nt^2}.$$

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Hence, by Hoeffding's inequality,

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- ▶ Let us replace $n = 500$ and $t = \frac{1}{10}$. We have

$$\Pr\left[|\nu - \mu| \leq \frac{1}{10}\right] \geq 1 - 2e^{-10} \approx 1.$$

- ▶ That is, “ ν learns μ ” is **probably and approximately correct (P.A.C.)**.
- ▶ Now, we turn to the general learning setting.

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Notations

- ▶ $\{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \mathcal{X}$ are samples.
- ▶ $\{y_1, \dots, y_n\} \subseteq \mathcal{Y}$ are corresponding labels generated by the target function g .
- ▶ $\mathcal{S} = \{\mathbf{x}_i\}_{i=1}^n \subseteq \mathcal{X}$ are training samples.
- ▶ **Binary case:** $y_i \in \{-1, +1\}$ and $\mathcal{H} \ni f_{\boldsymbol{\theta}} : \mathcal{X} \rightarrow \{-1, +1\}$.
- ▶ **Example:** Perceptron learning algorithm:

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = \text{sign}(\boldsymbol{\theta}^\top \mathbf{x}).$$

↪ We will focus on binary linear classification. It can be generalized to real-valued y (i.e., regression) with the same conclusion. However, it is technical, and they do not add to the insight of our analysis of the binary case.

Error Measure

The learning goal (analogous to $\nu \approx \mu$) is

$$f \approx g$$

How to measure such an approximation equality?

- Point-wise error measure:

$e(f(\mathbf{x}), g(\mathbf{x}))$ is small for all possible data $\mathbf{x}$

- Examples:

squared error: $e(f(\mathbf{x}), g(\mathbf{x})) = (f(\mathbf{x}) - g(\mathbf{x}))^2$

binary error: $e(f(\mathbf{x}), g(\mathbf{x})) = 1_{[f(\mathbf{x}) \neq g(\mathbf{x})]}$

The squared error measure is mainly used for regression, while the zero-one measure is tailored for classification.

In-sample Error versus Out-of-sample Error

- **In-sample Error:** Given a set of **training samples** $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$,

$$\text{Er}_{\text{in}} = \frac{1}{n} \sum_{i=1}^n e(f(\mathbf{x}_i), g(\mathbf{x}_i))$$

- **Out-of-sample Error:** Suppose data $\mathbf{x}$ follows a certain distribution $\mathcal{D}$ in an i.i.d. manner,

$$\text{Er}_{\text{out}} = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [e(f(\mathbf{x}), g(\mathbf{x}))]$$

Remarks:

- The In-sample error Er_{in} is also known as the **training error**.
- The out-of-sample error Er_{out} is **more general than the test error**.
Fortunately, we can use the test error to approximate Er_{out} very well when the test dataset is large enough.

Learning g is to Make Er_{out} Small

Recall that learning is all about to infer g outside of the seen training dataset, i.e.,

Make the out-of-sample error small

Final exam analogy

- ▶ The in-sample error/training error is the sample final.
- ▶ The out-of-sample error/test error is the actual final.
- ▶ Goal: do well on actual final.

Memorization vs learning

- ▶ Do well on training data by memorizing it (**overfitting**).
- ▶ Learning means you have to do well with new data (**generalization**).

But, Er_{out} is **not even computable** ;-)

The Concept of Training versus Testing / Generalization

The goal of generalization is to

explore how out-of-sample error is related to in-sample error

The reason to explore this relationship is that Er_{in} is computable, checkable, and even amenable (can be directly minimized).

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However, this intuition is not true in general.

The Fundamental Trade-off in Learning

Here is a simple decomposition:

$$Er_{\text{out}} = \underbrace{Er_{\text{out}} - Er_{\text{in}}}_{\text{generalization error}} + \underbrace{Er_{\text{in}}}_{\text{training error}}$$

Simple observation is that we have to simultaneously make generalization error and training error small, in order to make Er_{out} small.

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On the **generalization** side, we need:

less complex hypothesis $\mathcal{H}$

On the **training** side, we need:

more complex hypothesis $\mathcal{H}$

↪ We will study the above conclusions later.

i.i.d. Assumption

We make the following **assumption** to connect training and test data:

- ▶ All data (training + testing) come from the same distribution $\mathcal{D}$ (**identically distributed**).
- ▶ The data are sampled **independently**.

i.i.d. interpretation

	age	gender	salary	citizenship	years in job
applicant 1 (train)	2.5	1	10	3	1
applicant 2 (train)	2.8	0	8	6	5
applicant 3 (train)	1.6	0	0	4	0
applicant 4 (test)	1.5	1	4	5	3

- ▶ Rows 1-4 follow the same same distribution $\mathcal{D}$.
- ▶ Rows 1-4 are mutually independent.

i.i.d. is an ideal assumption, but a good approximation of practice.

The Starting Point

Given training samples $\mathcal{S} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$.

In expectation for fixed $f \in \mathcal{H}$: (we omit θ in f_θ for simplicity)

$$\mathbb{E}_{\mathcal{S} \sim i.i.d. \mathcal{D}} [\text{Er}_{\text{in}}(f)] = \text{Er}_{\text{out}}(f).$$

Proof: Recall that all samples are i.i.d. according to $\mathcal{D}$, we have

$$\begin{aligned} \mathbb{E}_{\mathcal{S} \sim i.i.d. \mathcal{D}} [\text{Er}_{\text{in}}(f)] &= \mathbb{E}_{\mathcal{S} \sim i.i.d. \mathcal{D}} \left[\frac{1}{n} \sum_{i=1}^n e(f(\mathbf{x}_i), g(\mathbf{x}_i)) \right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [e(f(\mathbf{x}), g(\mathbf{x}))] \quad (\text{since i.i.d.}) \\ &= \text{Er}_{\text{out}}(f) \end{aligned}$$

Interpretation

$$\mathbb{E}_{\mathcal{S} \sim i.i.d. \mathcal{D}} [\text{Er}_{\text{in}}(f)] = \text{Er}_{\text{out}}(f).$$

- ▶ $\text{Er}_{\text{in}}(f)$ is an **unbiased estimator** for $\text{Er}_{\text{out}}(f)$.
- ▶ Law of large numbers: When $n \rightarrow \infty$, we have in-sample error estimates the out-of-sample error accurately.
- ▶ However, this is an **asymptotic** (infinite n) result. In practice, we can never deal with infinite samples.

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Solution: Derive **non-asymptotic** (finite n) results using concentration inequalities. $\rightsquigarrow$ Next lecture.